

Product Requirements Document (PRD)

Live Event Gaming Platform (MVP)

Version: 1.0

Status: Approved for Development

1. Product Overview

A web-based platform that allows a host to run live interactive events where participants join through a shared link and compete in real time.

The initial release will support Quiz events. The platform will be designed to support additional game types in future releases, including puzzles, polls, treasure hunts, and custom challenges.

Participants can join from any browser without installing software.

2. Product Vision

Create a flexible live-event platform that enables organizations, teams, schools, communities, and event organizers to conduct engaging real-time competitions and activities.

The platform should support:

- Live quizzes
- Knowledge competitions
- Interactive games
- Team activities
- Future game formats

without requiring major changes to the core event workflow.

3. Goals

Primary Goals

- Allow hosts to create and run live quiz events.
- Allow participants to join through a shared link.
- Deliver questions in real time.
- Collect answers in real time.

- Display a live leaderboard.
 - Support up to 50 simultaneous participants.
 - Operate from a locally hosted environment using ngrok.
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Future Goals

- Support 100+ participants.
 - Cloud deployment.
 - Multiple game types.
 - Team competitions.
 - Analytics and reporting.
 - AI-assisted content generation.
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4. User Roles

Host

A user responsible for creating and managing events.

Host capabilities:

- Upload question banks
 - Create events
 - Configure event settings
 - Start event
 - Pause event
 - Resume event
 - Advance to next question
 - End event
 - Monitor participation
 - View leaderboard
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Participant

A user who joins and participates in an event.

Participant capabilities:

- Join event using link
- Enter display name
- Wait in lobby
- View questions
- Submit answers
- View leaderboard
- View final results

5. Supported Event Type (MVP)

Quiz

Supported question types:

Multiple Choice (MCQ)

Example:

What is the capital of Japan?

- Tokyo
- Seoul
- Bangkok
- Beijing

Text Answer

Example:

Who invented the telephone?

Number Answer

Example:

How many players are there in a cricket team?

True / False

Example:

The Earth revolves around the Sun.

6. Question Bank Management

Upload-Based System

Question banks are uploaded by hosts.

Supported formats:

- XLSX
- CSV

Manual question entry is not required in MVP.

Question Structure

Every question must contain:

- Question Type
 - Question Content
 - Correct Answer
 - Field / Category
 - Difficulty Level
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Field / Category

Represents the topic or domain of the question.

Examples:

- General Knowledge
- Technology
- Programming
- Cricket
- Sports
- Movies
- History
- Geography
- Science
- Company Knowledge

Purpose:

- Filter questions by category
 - Create category-specific events
 - Mix multiple categories
 - Support future random question generation
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Difficulty Level

Represents how difficult a question is.

Scale:

Level	Description
1	Very Easy
2	Easy
3	Beginner
4	Moderate
5	Intermediate
6	Challenging
7	Hard
8	Very Hard
9	Expert
10	Extreme

Purpose:

- Create beginner-friendly events
- Create expert-level competitions
- Filter questions by difficulty
- Support future adaptive quizzes

Question Bank Features

Hosts can:

- Upload question banks
- Preview uploaded content
- Validate uploaded files
- Reuse question banks across multiple events

Question banks contain only content.

Event behavior is configured separately.

7. Event Creation

When creating an event, the host configures:

Event Information

- Event Name
- Event Description

Question Source

- Select Question Bank
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Category Selection

Options:

- Use all categories
- Select specific categories

Examples:

- Technology only
 - Cricket only
 - Technology + Movies
 - All categories
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Difficulty Selection

Options:

- Use all difficulty levels
- Define minimum and maximum difficulty range

Examples:

- Difficulty 1–3
 - Difficulty 4–7
 - Difficulty 8–10
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Event Rules

Host configures:

- Time limit per question
- Base points per question
- Enable / Disable speed bonus
- Show leaderboard after each question
- Auto advance or manual advance

These settings belong to the event and not to individual questions.

8. Event Flow

Pre-Event Lobby

Participants:

- Join event
- Enter display name
- Wait in lobby

Host:

- Monitor participant count
 - Start event when ready
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During Event

For each question:

1. Question is displayed.
 2. Timer begins.
 3. Participants submit answers.
 4. Answers are locked when time expires.
 5. Scores are calculated.
 6. Leaderboard is updated.
 7. Host advances to next question.
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Event Completion

After the final question:

- Final leaderboard displayed
 - Winner announced
 - Event marked as completed
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9. Leaderboard

Leaderboard displays:

- Rank
- Participant Name
- Score

Leaderboard update modes:

- After every question
- Only at event completion

Configurable by host.

10. Real-Time Requirements

The platform must support:

- Real-time participant joining
- Real-time question delivery
- Real-time answer submission
- Real-time leaderboard updates
- Real-time event state changes

All participants should experience synchronized gameplay.

11. Host Control Panel

The host dashboard must provide:

Event Controls

- Start Event
 - Pause Event
 - Resume Event
 - Next Question
 - End Event
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Monitoring

- Participant Count
 - Current Question
 - Submitted Answers Count
 - Live Leaderboard
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12. Participant Experience

Join Screen

Participants can:

- Open event link
 - Enter display name
 - Join event
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Lobby Screen

Displays:

- Event information
 - Waiting status
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Question Screen

Displays:

- Current question
 - Available answers/options
 - Remaining time
 - Submit button
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Leaderboard Screen

Displays:

- Current ranking
 - Current score
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Results Screen

Displays:

- Final ranking
 - Final score
 - Winner information
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13. Non-Functional Requirements

Performance

- Support up to 50 concurrent participants.
 - Real-time updates should feel instantaneous.
 - Event progression should remain synchronized.
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Compatibility

Supported devices:

- Desktop
- Mobile
- Tablet

Supported browsers:

- Chrome
 - Edge
 - Firefox
 - Safari
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Installation

Participants should not install any software.

Browser-only experience.

14. MVP Scope

Included:

- Quiz Events
- XLSX Upload
- CSV Upload
- MCQ Questions
- Text Questions
- Number Questions
- True / False Questions
- Question Categories
- Question Difficulty Levels
- Event Configuration
- Real-Time Gameplay

- Live Leaderboard
 - Host Controls
 - ngrok-Based Access
 - Support for up to 50 participants
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Not Included:

- Team Mode
 - Chat System
 - AI Question Generation
 - Analytics Dashboard
 - Puzzle Games
 - Poll Events
 - Treasure Hunts
 - Authentication System
 - Cloud Deployment
 - Public Events Directory
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15. Future Roadmap

Phase 2

- AWS Deployment
 - 100+ Participants
 - Team Competitions
 - Analytics
 - Event Reports
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Phase 3

- Puzzle Events
 - Poll Events
 - Treasure Hunt Events
 - Custom Game Engine
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Phase 4

- AI Question Generation
 - Tournament Mode
 - Multi-Host Events
 - Public Event Catalog
 - Advanced Event Automation
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Success Criteria

A host can:

1. Upload a question bank.
2. Create an event.
3. Select categories and difficulty levels.
4. Configure event rules.
5. Share an event link.
6. Allow participants to join.
7. Conduct a complete live quiz.
8. Display a final leaderboard.

All within a browser-based experience supporting up to 50 concurrent participants.